

RISHAB MURUGESAN

Robotics Engineer | System Administrator | Freelancer

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CAREER OBJECTIVE :

Motivated Robotics and Automation engineering graduate with strong foundations in ROS2, mechatronics, embedded systems and automation. Seeking an opportunity to apply theoretical knowledge to real-world robotics projects and contribute to innovative engineering solutions in a growth-oriented organization.

EDUCATION CREDENTIALS :

2022 - 2026 | Erode Sengunthar Engineering College - Erode.

Bachelor of Engineering - Robotics and Automation.

CGPA : 8.92.

2022 | Ayyanar Matric.Hr.Sec School - Veppur.

Higher Secondary Certificate (HSC) - Computer Science.

Marks: 485 / 600 (80%).

2020 | Ayyanar Matric.Hr.Sec School - Veppur.

Secondary School Leaving Certificate (SSLC).

Marks: 485 / 500 (97%).

TECHNICAL SKILLS :

- ✓ ROS2 (Robot Operating System).
- ✓ Automation & IoT Integration.
- ✓ Python Programming.
- ✓ Rapid Prototyping.
- ✓ 3D Modeling (Tinkercad)
- ✓ CAD Modelling (SolidWorks) Basics.
- ✓ Drone Developing & Piloting Basics.
- ✓ Git & Github.

AREA OF INTEREST :

- ✓ Industrial Robotics & IoT.
- ✓ Robotics Trainer.
- ✓ Research and Development.
- ✓ Supervisor and Quality Engineer

MS OFFICE TOOLS :

- ✓ Microsoft Word.
- ✓ Microsoft Excel.
- ✓ Microsoft PowerPoint.

PROJECTS :

Agriculture Robot.

Smart Harvesting Robot | 05 Dec 2024 - 10 Feb 2025.

- AI-enabled robot detects and harvests ripe crops with high precision.
- Navigates autonomously while monitoring crop health using IoT sensors.

Industrial Robot.

SCARA Robot | 03 Mar 2025 - 20 May 2025.

- SCARA robot enables fast and precise lifting of heavy industrial loads.
- Its rigid arm design ensures accurate and stable horizontal movement.
- Multi Mode Control (Automation, Semi Automation (GUI), Manual Control).

Road Inspection Sytem using UAV

ROS2 Based Road Inspection Sytem using UAV | 22 Jan 2026 - 13 March 2026.

- Developed an autonomous UAV-based road inspection system using ROS2 for real-time road condition monitoring.
- Implemented YOLOv8 for accurate pothole detection from aerial images.
- Integrated Raspberry Pi 4 and NEO-6M GPS for edge processing, geospatial mapping and location-based pothole reporting.

EXPERIENCE :

Internship

Company : Codec Technologies

Robotics Intern | Remote | 20 Feb 2025 - 20 Mar 2025.

- Learned basics of ROS and industrial automation.
- Developed simple robot projects like Maze solver, line following robot.

Graduate Trainee (Quality)

Company : ZF CVCS India Ltd, Ambattur- - Chennai.

Trainee (Quality Engineer) | Hybrid | 01 Nov 2025 - 15 Dec 2025.

- Trained in automotive manufacturing and quality processes at ZF CVCS, Ambattur.
- Gained practical experience in inspections, reporting and supporting production teams.

CERTIFICATIONS :

✓ PMGDISHA Course Completed

Appreciation of Digital literacy | 19 Sep 2022.

✓ Skill Development Workshop

Topic: Computer Vision / LLM | 26 Sep 2023.

Organizer : Chennai Institute of Technology - Chennai.

✓ Drone Development Workshop

Organizer : ESEC Incubation & Garudan Unmanned System Pvt.Ltd - Coimbatore.

Tenure : 25 April 2024 & 26 April 2024.

✓ Robotics Technology

Company : Aero Knotz Drones India Pvt.Ltd - Chennai.

Robotics and Basics of Drone | 24 Sep 2024 - 26 Sep 2025.

✓ Skill Development Course

Topic: Rapid Prototyping Techniques | 10 Feb 2025 to 21 Feb 2025.

Organizer : ESEC - Skill Training and Innovation Center.

✓ Python Programming Course

Platform : Udemy Online Courses.

✓ ROS2 BOOTCAMP

Organizer : VIT- Chennai | 29 Oct 2025 & 30 Oct 2025.

ACHIEVEMENTS :

✓ BEST PROJECT

Organizer : Student Confluence - EXCEL Engineering college - Salem.

Event : Project Expo | 09 April 2025.

✓ BEST HARDWARE PROJECT

Organizer : IEEE Industrial Electronics & KPR Institutions - Coimbatore.

Event : Project Expo | 30 April 2025.

✓ **BEST EMBEDDED SOLUTION AWARD**

Organizer : TNSKILL & Naan Mudhalvan Scheme by the Tamilnadu Government.

Event : National level Hackathon | 09 May 2025.

Theme : Low-Cost, High-Efficiency Embedded & IOT Solutions for SDG Goals.

✓ **GOLD WINNER**

Organizer : Drone Federation of India & Meity.

Event : NIDAR - National Innovation Challenge for Drone application & Research.

Theme : Precision Agriculture.

Role : ROS Developer & Testing Engineer.

Tenure : 10 Jan 2026 to 16 Jan 2026.

AWARDS :

✓ **BEST OUTGOING STUDENT AWARD**

College : Erode Sengunthar Engineering College, Erode.

Academic Year : 2022 -2026.

✓ **STUDENT ACHIEVER AWARD**

College : Erode Sengunthar Engineering College, Erode.

Event : Drone Innovation Challenge - NIDAR Event.

Academic Year : 2025 - 2026.

LEADERSHIP EXPERIENCE :

TREASURER

Rotaract Club : Erode Sengunthar Engineering College - Erode.

Tenure : May 2024 - April 2025.

PRESIDENT

RoboTrix Association : Erode Sengunthar Engineering College - Erode.

Tenure : August 2025 - May 2026.

SOFT SKILLS :

- ✓ Logical Thinking.
- ✓ Communication.
- ✓ Problem Solving.
- ✓ Time Management.
- ✓ Adaptability.

LANGUAGES KNOWN :

- ✓ **English (Fluent).**
Read,Write,Speak.
- ✓ **Tamil (Native).**
Read,Write,Speak.
- ✓ **Hindi**
Speak.

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge and belief.


RISHAB MURUGESAN